

Siddhartha Manu

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EDUCATION

KIET Group of Institutions, B.Tech. IT

CGPA 7.55

Ghaziabad, Uttar Pradesh

Nov. 2022 – Jun 2026

Jagran Public School, 12th CBSE

Percentage 90.6%

Lucknow, Uttar Pradesh

April 2020 – March 2021

PROJECTS

Handwritten Digit Classifier | *Pytorch, Numpy, Tkinter, Matplotlib*

May 2026 - Present

- Designed and Trained a modular Multi-Layer Perceptron (MLP) for classification using a filtered *MNIST* dataset.
- Built and Interactive GUI supporting real-time digit drawing and **confidence-based inference**
- Implemented a custom pipeline including **cropping, centering, normalization and aspect-ratio preservation**
- Integrated different optimizers such as **SGD, Momentum, and ADAM** to check which gives the maximum accuracy and use it accordingly

URL Shortner | *Fast API, Pydantics, SQLite, SQLAlchemy*

Dec 2025 - Jan 2026

- Built a RESTful FastAPI backend for a multi-user URL Shortening service, with Create, Read, Update, and Delete operations for URLs
- Implemented **collision - resistant** short code generation ensuring one-to-one URL mappings.
- Designed **Pydantic schemas and SQLAlchemy** in order to make sure sensitive details are not accessible to anyone in the system

Even Steven: Regularization Analysis | *PyTorch, Matplotlib, Numpy*

Jun 2026

- Built a configurable Multi-Layer Perceptron(MLP) framework for image classification on the **Google Quick, Draw!** dataset
- Implemented and evaluated multiple regularization techniques including **Dropout, L2 Weight Decay, Batch Normalization** and **Data Augmentation**
- Analyzed training and validation curves to identify overfitting and demonstrated improved generalization, increasing test accuracy from **85.04%** to **87.14%** through regularization

TECHNICAL SKILLS

Languages and Frameworks: Python, C++(Data Structures and Algorithms), SQL (Postgres), PyTorch (Deep Learning), Scikit-Learn(Machine Learning), OpenGL, FastAPI

Tools and Libraries: Git, Visual Studio Code, Visual Studio, PyCharm, LINUX, NumPy, Matplotlib, Pydantic, Jupyter Notebook, SQLAlchemy, dotenv, Pickle

POSITION OF RESPONSIBILITY

Student Induction Programme|*Instructor*

Sep 2024

- Gave Students a brief introduction to Data Structure and Algorithms with a strength of over **150 students**
- Gave students a brief introduction to Machine Learning and the Maths behind it, with a strength of over **50 students**

Virtual Vortex (Amiphoria 2024)|*Host*

Mar 2024

- Successfully hosted a multiplayer gaming event of a tactical FPS Valorant with a footfall of **more than 100 participants**

RESEARCH

Solar Energy Assessment Through BIPVs| *Research Paper*

May 2025 - Present

- Performed analysis of how BIPVs can complement solar energy generation when used with BAPVs.
- It uses LOD 1 3D city model in order to see how the buildings are oriented and how can we optimize the placement of BIPVs in order to get the maximum possible output.